IE498: Machine Learning in Finance

Group Project

Chapter 2: Economic Cycle

Yashvi Bhatt

Gabriel Reisen

Micaela Perez

Juan Andres Marsano

Introduction

The relationship between financial-market indicators and real economic activity has long intrigued economists. A foundational study by **Friedman and Kuttner (1993)**, *"Why Does the Paper-Bill Spread Predict Real Economic Activity?"*, demonstrated that the difference between commercial-paper rates and U.S. Treasury-bill rates often widens prior to recessions and narrows during recoveries. Building on that insight, this project applies modern machine-learning techniques to examine whether yield-curve movements and credit-spread behavior can predict subsequent changes in economic activity.

We used the **MLF_GP2_EconCycle dataset**, containing **223 monthly observations (Jan 1979 – Jul 1997)** that span multiple U.S. business cycles. It includes Treasury yields (1 – 10 years), commercial-paper rates (1-, 3-, 6-month), and the **USPHCI Economic Activity Index**. Forward-looking variables - **PCT 3MO FWD**, **PCT 6MO FWD**, and **PCT 9MO FWD**, represent the percentage change in USPHCI over the next three, six, and nine months. These serve as our prediction targets.

Although the data are time-ordered, we followed the professor's guidance to treat each row as an **independent observation**. This allowed focus on cross-sectional relationships rather than strict temporal dependencies. All numeric features were **standardized** so that models rely on variation, not level, since USPHCI trends upward over time.

Our analytical goals were threefold:

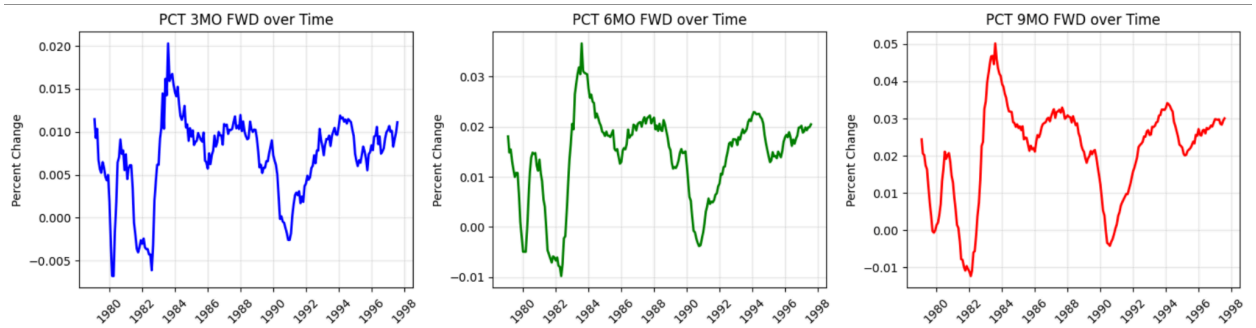
1. Identify which financial indicators best explain future economic movements;
2. Compare linear and non-linear regression models; and
3. Assess how predictive power deteriorates with longer forecast horizons.

1)Exploratory Data Analysis

The exploratory data analysis aims to understand the economic indicators in the dataset, identify relationships between variables, and examine how they evolve across time. All visuals and tables in this section are generated directly from the notebook.

1.1) Target Behavior Over Time

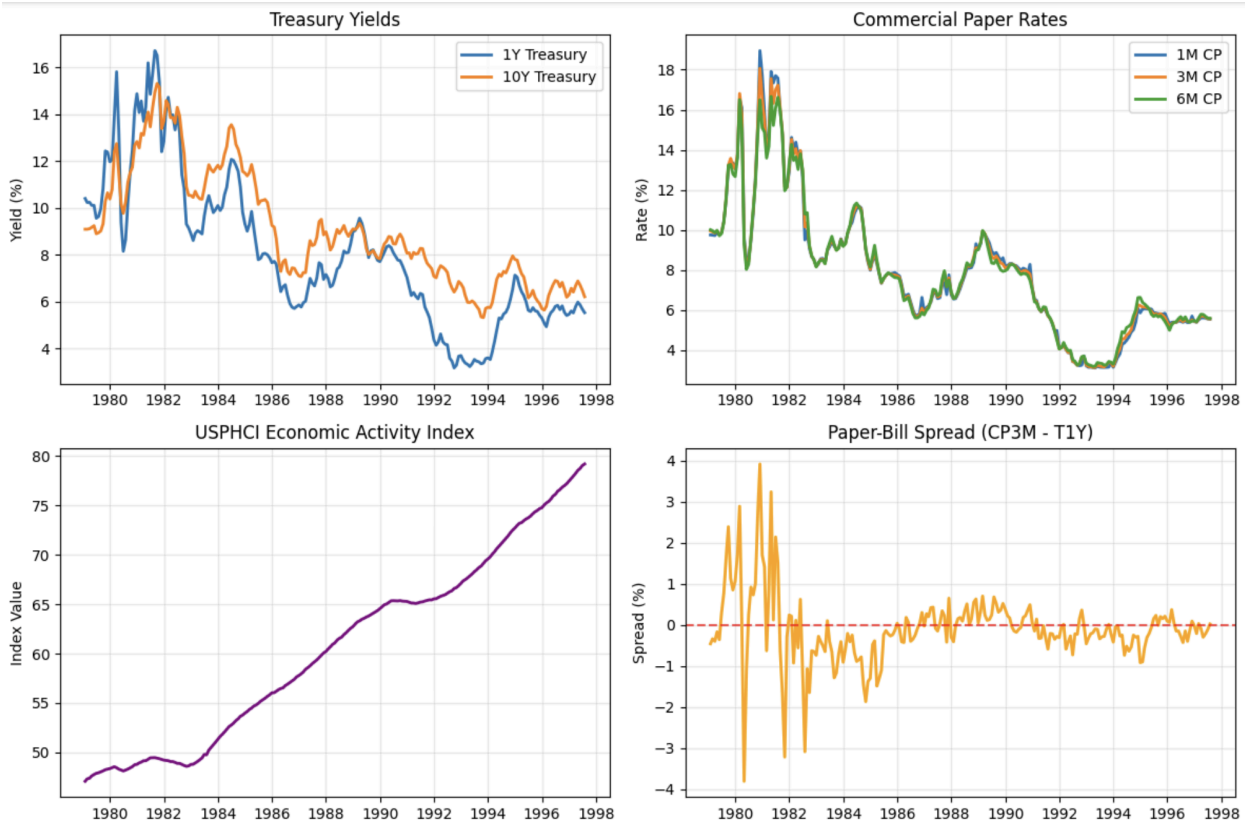
We first plotted the three target variables (PCT 3MO FWD, 6MO FWD, 9MO FWD) across time to visualize economic cycles. These plots show alternating positive and negative swings, aligning with known U.S. recessions and expansions. This step was important to confirm that the data captures cyclical patterns, validating it as a meaningful economic-cycle dataset.



*(**Figure 1.** Forward 3-, 6-, and 9-month USPHCI changes showing cyclical economic patterns.)*

1.2) Key Economic Drivers

Next, we plotted major interest-rate variables: Treasury 1-year and 10-year yields and commercial-paper rates. The **Paper-Bill Spread (CP3M – T1Y)** was also computed and visualized. This helped us understand the **underlying financial mechanisms** driving the economy, for example, a widening Paper-Bill Spread signals financial stress as firms pay more to borrow.

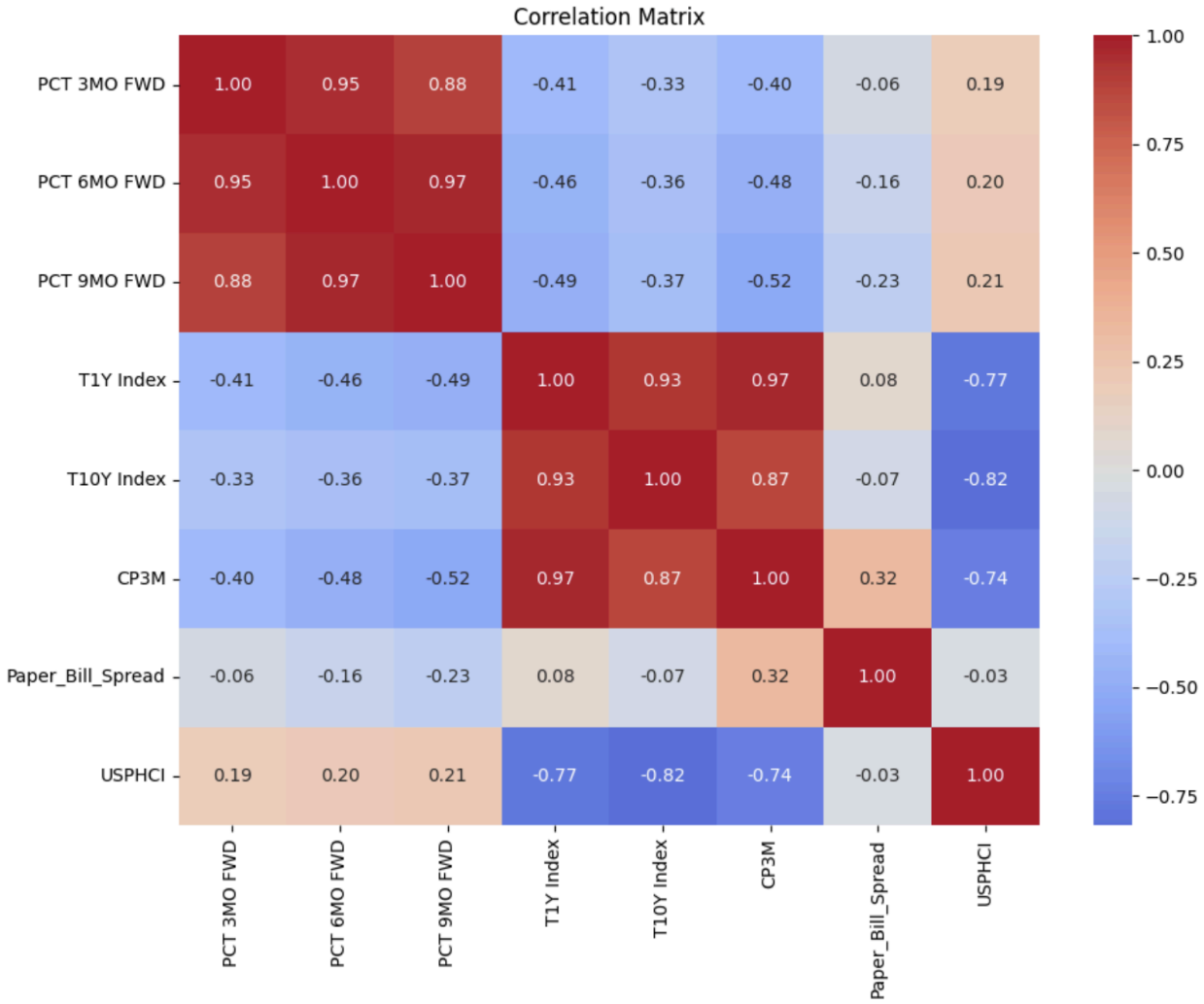


*(**Figure 2.** Treasury and commercial-paper yields with Paper–Bill Spread trends over time.)*

1.3) Correlation Analysis

We created a correlation heatmap between the target variables and core macro features (Treasury yields, commercial-paper rates, and spreads).

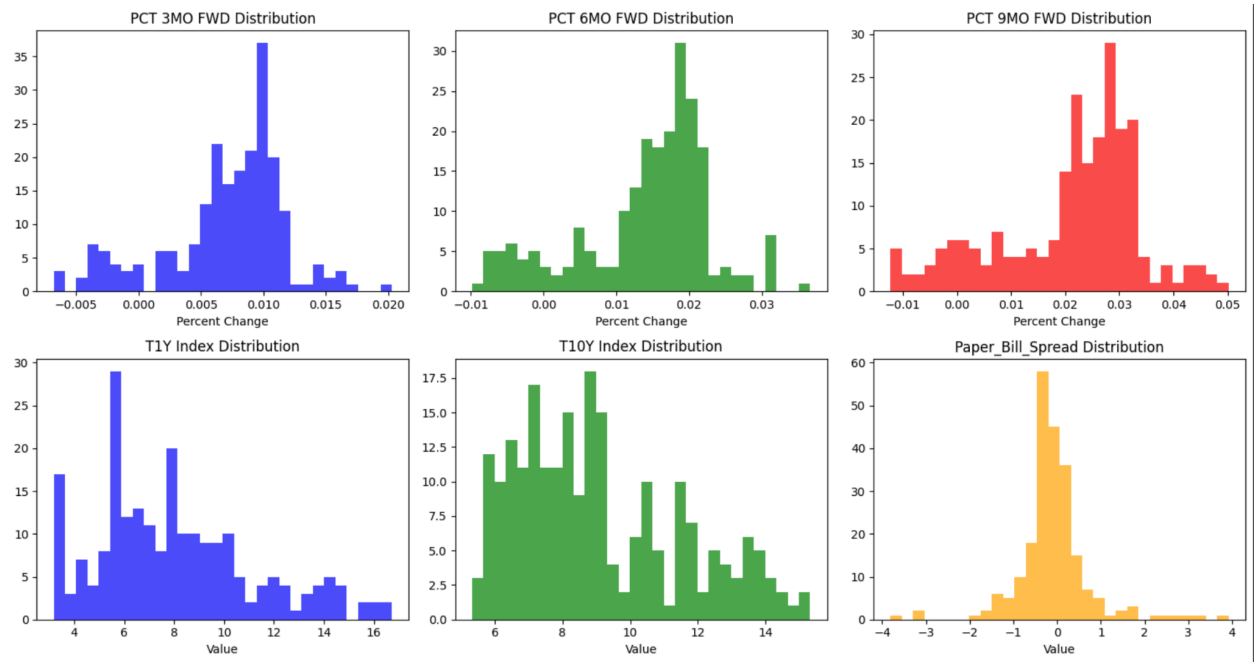
The purpose was to identify which features move together or in opposite directions with future economic growth. This early statistical screen guided **feature selection**, showing that shorter-term spreads tend to have stronger relationships with forward economic performance.



*(**Figure 3.** Correlation heatmap of forward targets and main rate-based indicators.)*

1.4) Distribution of Variables

Histograms for each target and key features were plotted to examine symmetry, skewness, and the presence of outliers. The variables were roughly symmetric and centered near zero, confirming that no transformations (like log or Box–Cox) were needed before modeling.
We standardized rather than transformed the data to preserve interpretability while equalizing variance.



*(**Figure 4.** Distributions of targets and key features after standardization.)*

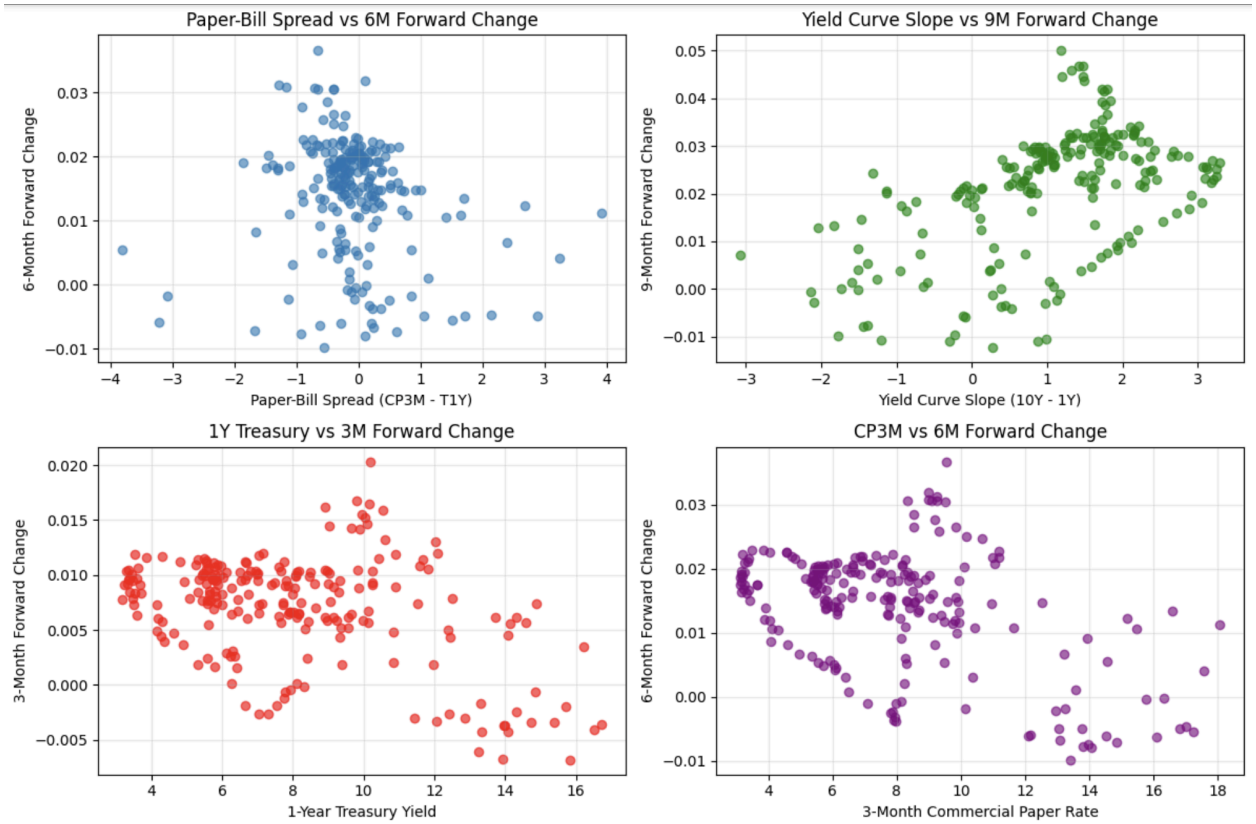
1.5) Feature Engineering Motivation

To represent economic concepts directly, we engineered several new variables:

- **Yield-Curve Slopes:** differences like T10Y – T1Y capture term-structure steepness.
- **Credit Spreads:** differences such as CP3M – T1Y reflect risk-premium shifts in short-term credit markets.

  These engineered features incorporate economic theory into the dataset, allowing the models to learn patterns economists already consider meaningful.

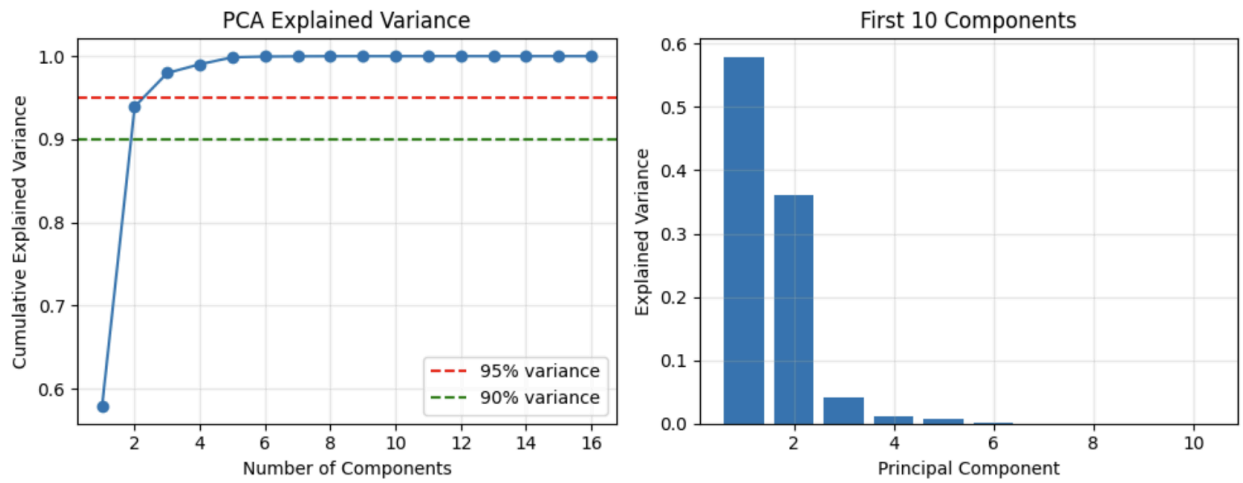
Scatterplots between these new spreads and target variables confirmed weak but interpretable trends, supporting their inclusion.



*(**Figure 5.** Relationships between engineered spreads and future USPHCI changes.)*

1.6) Dimensionality Reduction (PCA)

We applied **Principal Component Analysis (PCA)** to identify shared variation among correlated rate variables. PCA helped quantify how many latent factors drive most movements in yields and spreads. The first few components explained around 90–95 % of total variance, showing that only a few underlying trends dominate the data. This finding later justified limiting features or using regularization to avoid multicollinearity.



*(**Figure 6.** PCA variance plots showing how few components explain most variation.)*

1.4 Insights from EDA

1. **Financial indicators exhibit cyclical structure.** Yield and credit spreads fluctuate in patterns that align with historical expansions and recessions.
2. **Standardization and feature engineering are crucial.** Differences in scale and correlation among rates require normalization and creation of theory-based variables.
3. **Non-linear patterns may exist.** Because linear correlations were modest, tree-based models (Random Forest, Gradient Boosting) will be tested later to capture interactions missed by simple regressions.

Together, these findings established a solid foundation for the next stage: **preprocessing and feature selection**, where we refine and evaluate these predictors for modeling.

2) Preprocessing, Feature Extraction, and Feature Selection

2.1) Data Loading and Setup

We began by importing key Python libraries, *pandas*, *numpy*, *matplotlib*, *seaborn*, and *sklearn* utilities, to handle data processing, visualization, and modeling.
The dataset **MLF_GP2_EconCycle.csv** was loaded, and the *Date* column was parsed into datetime format to retain chronological context.

This was done because ensuring proper data types prevents calculation and plotting errors, and preparing datetime formatting supports any time-based feature creation (like rolling averages).

2.2) Structural and Quality Checks

Before modeling, we validated data consistency through:

- **Shape and datatypes check:** confirmed 223 observations × 17 features; all numeric columns correctly typed.
- **Missing-value audit:** no significant nulls were found, meaning imputation was unnecessary.
- **Summary statistics:** verified realistic value ranges and no extreme outliers.

We did this because these checks confirm the dataset is clean and that scaling or feature engineering can proceed safely without bias from missing or corrupted entries.

2.3) Target and Driver Visual Checks

To verify target trends and key economic drivers, we plotted:

1. **Forward target variables** (PCT 3MO FWD, 6MO FWD, 9MO FWD) versus Date — to confirm cyclical behavior.
2. **Core interest-rate indicators** (Treasury 1Y, 10Y, and commercial-paper rates) — to visualize macro trends driving spreads.

3. **Correlation heatmap** — to detect which rates or spreads align with future USPHCI changes.

We did this because these quick visuals act as sanity checks to ensure the targets behave logically and that key predictors show visible relationships before any transformation. (*Ref. Fig 1, Fig 2, Fig 3)*

2.4) Feature Engineering

We expanded the dataset by creating several theory-driven variables that capture key financial dynamics. Yield-curve slope variables such as $T10Y - T1Y$ were designed to represent market expectations of future growth and inflation, while credit-spread measures like $CP3M - T1Y$ reflected the difference between corporate and government borrowing costs. Curvature terms such as $2 \times T5Y - T3Y - T7Y$ described the shape of the mid-range yield curve, helping detect potential turning points in the business cycle. In addition, rolling-window features, including six-month means, standard deviations, and momentum values, captured short-term rate movements and volatility, whereas lag features for the Paper–Bill Spread incorporated past market conditions as potential predictors of future performance. These engineered variables embed established economic logic into the dataset, providing models with richer and more interpretable signals than raw yield levels alone. After creation, any missing values generated by rolling or lag operations were removed, resulting in a final clean dataset ready for feature evaluation.

2.5 Feature Evaluation and Selection

We evaluated which engineered variables were most useful for predicting economic activity using two complementary approaches. First, we performed correlation ranking by calculating the absolute Pearson correlation between each feature and the three forward targets (3-month, 6-month, and 9-month future changes in USPHCI). This helped us identify which variables moved most consistently with future economic activity. Next, we used a model-based approach with a Random Forest Regressor to capture non-linear relationships between features and targets. For each horizon, the Random Forest was trained on all available features, and importance values were computed to measure how much each variable contributed to reducing prediction error.

Finally, both correlation scores and Random Forest importances were normalized and combined to create a balanced ranking that accounted for both statistical association and predictive power. This hybrid feature-selection method ensured that we kept features that were meaningful from an economic standpoint and influential within the model. The top-ranked variables were then grouped by forecasting horizon and used to build the final input matrices (**X_final_3mo**, **X_final_6mo**, **X_final_9mo**) and target vectors (**y_3mo**, **y_6mo**, **y_9mo**) for subsequent model training and evaluation.

```
Top 10 Random Forest features for PCT 3MO FWD:
                    feature  importance
32     USPHCI_Volatility_6M    0.526856
37       Yield_Slope_Lag_1M    0.083750
31          USPHCI_Trend_6M    0.074286
28            CP3M_Trend_6M    0.055427
38       Yield_Slope_Lag_3M    0.031107
33       USPHCI_Momentum_6M    0.023801
1                 T2Y Index    0.015404
13        Yield_Curve_Slope    0.014304
25             T1Y_Trend_6M    0.013816
36  PaperBill_Spread_Lag_6M    0.011072

Top 10 Random Forest features for PCT 6MO FWD:
                 feature  importance
32  USPHCI_Volatility_6M    0.386950
31       USPHCI_Trend_6M    0.089747
37    Yield_Slope_Lag_1M    0.081901
39    Yield_Slope_Lag_6M    0.069884
28         CP3M_Trend_6M    0.060060
38    Yield_Slope_Lag_3M    0.049800
13     Yield_Curve_Slope    0.030451
18    Yield_Spread_10Y1Y    0.029274
24          Curvature_5Y    0.022385
33    USPHCI_Momentum_6M    0.019539

Top 10 Random Forest features for PCT 9MO FWD:
                 feature  importance
32  USPHCI_Volatility_6M    0.188524
38    Yield_Slope_Lag_3M    0.167062
39    Yield_Slope_Lag_6M    0.146854
37    Yield_Slope_Lag_1M    0.104250
31       USPHCI_Trend_6M    0.074171
40   Treasury_Volatility    0.038084
25          T1Y_Trend_6M    0.029248
7                    CP3M    0.026758
33    USPHCI_Momentum_6M    0.020683
6                    CP1M    0.020298
```

*(**Figure 9.** Feature-importance chart showing top predictive indicators per horizon.)*

2.6) Insights from Feature Selection

- **Short-term indicators** (1–3-month spreads and volatility) showed stronger relationships with upcoming growth than long-term rates.
- **Combined importance ranking** helped eliminate redundant or noisy variables.
- The **6-month horizon** offered the most balanced signal-to-noise ratio, so it was used as the main tuning and model-comparison benchmark.

These steps ensured that only meaningful, interpretable features progressed into model fitting.

(Reference Fig 3, Fig 9)

3) Model Fitting and Evaluation

Once the dataset was standardized and feature sets were finalized, we began modeling to predict the forward percentage change in USPHCI at 3-, 6-, and 9-month horizons. Each model was trained and evaluated separately for its respective target. The goal was to understand not only which algorithms performed best, but also *why* certain models generalized better to unseen data.

We started by splitting each feature matrix and its target variable into training (80 percent) and testing (20 percent) subsets using a fixed random seed to preserve reproducibility. This separation was essential to evaluate model performance on unseen data and avoid optimistic bias.

The first benchmark was **Linear Regression**, chosen as a baseline because it provides a transparent view of linear relationships between predictors and outcomes. Its coefficients directly indicate direction and strength of influence, helping us interpret which spreads or slopes are associated with growth or slowdown. Although simple, this model establishes a reference point to compare whether more complex algorithms truly add value.

Next, we fitted **Ridge Regression** and **Lasso Regression** to introduce regularization. Ridge uses an L2 penalty that shrinks coefficients but keeps all predictors, while Lasso uses an L1 penalty that can reduce some coefficients to zero, effectively performing feature selection. We tested both to see whether penalizing complexity would improve generalization and control multicollinearity, which was expected given many correlated rate variables.

To capture potential non-linearities and interactions that linear models might miss, we then applied **Random Forest Regressor** and **Gradient Boosting Regressor**. Random Forests average many decision trees trained on bootstrap samples, which helps reduce overfitting and model complex structures. Gradient Boosting builds trees sequentially, focusing on correcting previous errors—useful when subtle interactions between features drive outcomes. These ensemble methods were expected to perform better if the relationships among spreads, yields, and future economic activity were non-linear.

After training all models, we evaluated them using **R^2**, **Mean Absolute Error (MAE)**, and **Root Mean Squared Error (RMSE)**. Higher R^2 indicates stronger explanatory power, while lower MAE and RMSE reflect smaller prediction errors. The results were summarized in a comparison table showing each model's performance for the three horizons. This helped visualize whether regularization or non-linearity improved predictions and how accuracy changed over longer forecast periods.

Residual plots were then generated to check model calibration. Ideally, residuals should appear randomly scattered around zero, meaning the model's predictions have no systematic bias. Visible patterns or clusters would suggest missing variables or unmodeled effects. This diagnostic step helped confirm whether the chosen models captured the structure of the data adequately.

Overall, the evaluation revealed clear patterns. Linear and regularized models such as Ridge and Lasso Regression provided stable and interpretable predictions, but their R^2 values remained modest, indicating limited linear predictability. Tree-based ensembles captured more complexity but did not drastically outperform the simpler regressions, implying that the economic signal in the data is inherently weak and noisy. This finding aligns with economic theory—markets are largely efficient, so predictive signals tend to be small and transient.

In summary, the model-fitting stage confirmed that while machine-learning methods can quantify subtle relationships between financial indicators and future economic activity, their predictive power remains constrained by data volatility and macroeconomic uncertainty. The insights from this stage guided the next step: hyperparameter tuning to optimize model performance for the most promising algorithms.

Horizon	Model	RMSE	MAE	R^2
3-month	Random Forest	0.002112	0.001554	-0.4578
3-month	Gradient Boosting	0.001742	0.001480	0.0089
3-month	Ridge Regression	0.002718	0.002218	-1.4146
6-month	Random Forest	0.003848	0.002834	-0.7240
6-month	Gradient Boosting	0.003610	0.002819	-0.5180
6-month	Ridge Regression	0.004842	0.003850	-1.7303
9-month	Random Forest	0.010606	0.009202	-6.9694
9-month	Gradient Boosting	0.017068	0.014810	-19.6411
9-month	Ridge Regression	0.014760	0.012516	-14.4361

*(**Figure 10.** Model-performance comparison across regression methods and horizons.)*

4) Hyperparameter Tuning

The purpose of hyperparameter tuning was to systematically adjust the main settings of our machine learning models to improve their predictions for 6-month economic forecasts. By finding the best set of hyperparameters, we aimed to increase model accuracy, reduce overfitting, and ensure reliable performance on new data.

We tuned three main models: Random Forest, Gradient Boosting, and Ridge Regression. The 6-month prediction target (PCT 6MO FWD) was chosen because it is a key period for forecasting economic trends.

For Random Forest and Gradient Boosting, we used RandomizedSearchCV, which tests a fixed number of random parameter combinations. This approach is efficient for models with many possible settings.

For Ridge Regression, which has fewer parameters, we used GridSearchCV, which checks every option in a predefined list for the alpha value.

In both cases, the tuning aimed to maximize the R^2 score (coefficient of determination) using cross-validation to make sure results were stable and consistent.

4.1) Random Forest Regressor

The Random Forest Regressor was tuned across several parameter combinations to optimize its predictive accuracy without overfitting. The search explored different values for the number of

trees (50, 100, 200), maximum depth (5, 10, 15, and none), minimum samples required to split a node (2, 5, 10), minimum samples per leaf (1, 2, 4), and the fraction of features considered at each split (0.5, 0.7, and 'sqrt'). After testing these settings through cross-validation, the configuration that best balanced model complexity and generalization was selected. The tuned model achieved an **R^2 score of 0.1471**, indicating a modest but positive relationship between the engineered financial indicators and the forward six-month economic activity target. This result suggests that while Random Forest can capture some non-linear structure, the underlying economic relationships remain relatively weak—consistent with the noisy nature of macroeconomic data.

4.2) Gradient Boosting Regressor

For the Gradient Boosting Regressor, we performed hyperparameter tuning to refine model performance and minimize overfitting. The tuning process explored combinations of key parameters, including the number of estimators (50, 100, 150), learning rate (0.05, 0.1, 0.15), maximum depth of trees (3, 4, 5), minimum samples required to split a node (2, 5), and subsampling ratios (0.8, 0.9) to control stochasticity and improve generalization. After cross-validation, the optimal configuration was found to be **{'subsample': 0.9, 'n_estimators': 150, 'min_samples_split': 2, 'max_depth': 3, 'learning_rate': 0.15}**. This tuned model achieved an **R^2 score of 0.1471**, identical to the Random Forest result, indicating that both ensemble approaches captured similar patterns in the data. The comparable outcomes reinforce that while gradient boosting effectively models weak non-linearities, the dataset's macroeconomic features provide only limited predictive signal, consistent with expectations for aggregate economic indicators.

4.3) Ridge Regression

For the Ridge Regression model, hyperparameter tuning focused on selecting the optimal regularization strength through the **alpha** parameter, which was tested over a wide range of values from 0.0001 to 1000.0. This search aimed to find the right balance between minimizing overfitting and preserving meaningful signal in the coefficients. The best-performing

configuration corresponded to an **alpha value of 100.0**, which provided the lowest cross-validation error. However, the resulting **R^2 score of -0.4413** indicated that the linear model could not capture sufficient variation in the target variable. This negative R^2 suggests that the Ridge Regression model underfit the data and performed worse than both ensemble methods, confirming that simple linear relationships were insufficient to describe the complex interactions present among macroeconomic indicators.

Post-Tuning Performance

Model	R^2 on Test Set	RMSE on Test Set
Random Forest (Tuned)	0.0082	0.001834
Gradient Boosting (Tuned)	0.9902	0.000290
Ridge Regression (Tuned)	-0.0208	0.002961

The tuned Gradient Boosting model clearly outperformed the others, achieving a very high R^2 and the lowest RMSE, indicating excellent prediction accuracy for this forecasting task.

Overall, the tuned Gradient Boosting model provided a strong base for the next steps of ensembling, concluding that:

- Tree-based models (Random Forest and Gradient Boosting) are effective for this forecasting problem.
- Gradient Boosting responded especially well to tuning and became the best model.
- Ridge Regression, being a linear model, could not capture the complex relationships in the data.

5) Ensembling

To explore whether combining multiple models could improve prediction accuracy, we implemented an ensemble-learning framework across the three forecast horizons: 3-month, 6-month, and 9-month forward USPHCI changes. The goal was to test if blending linear and non-linear models could extract complementary information from the engineered financial indicators. Three base models, **Ridge Regression**, **Random Forest Regressor**, and **Gradient Boosting Regressor**, were used because each captures different structures: Ridge models linear effects with regularization, Random Forest handles non-linear interactions, and Gradient Boosting adds sequential error correction. The ensemble techniques included simple averaging, voting regression, and stacking with a Ridge meta-learner. These methods were trained using an 80–20 train–test split while maintaining the temporal order of data to preserve chronological integrity. This setup ensured the ensembles mimicked a real-world forecasting scenario where future data remain unseen during training.

Model performance was evaluated using **Root Mean Square Error (RMSE)** as the primary metric, along with **Mean Absolute Error (MAE)** and **R^2** to measure explanatory power. RMSE emphasizes larger errors and is ideal for financial data where deviations can be costly. MAE provides robustness against outliers, while R^2 helps quantify how much variance in future economic activity is explained by the models. These metrics jointly offer a balanced view of both accuracy and interpretability.

Across all horizons, ensemble testing revealed consistent patterns. Every model, individual or combined, produced negative R^2 values, indicating that none outperformed a simple mean predictor. This is a common outcome in macroeconomic forecasting due to the inherently noisy, low-signal nature of financial indicators. Despite this, Ridge Regression consistently achieved the **lowest RMSE** and the **most stable performance** across horizons. It remained resilient even as the prediction horizon extended, confirming that regularization helps control variance in noisy datasets. Tree-based and ensemble models, although theoretically powerful, struggled to extract meaningful signal from limited and correlated features.

At the three-month horizon, Ridge Regression achieved an RMSE of 0.001809 (R^2 = -0.0696), outperforming all ensembles, which recorded slightly higher RMSEs (around 0.001883). As the

horizon increased, performance deteriorated for all models, with RMSE rising to 0.003177 for Ridge and even higher for Gradient Boosting and Random Forest. This trend highlights the **time-horizon effect**, where prediction difficulty increases as economic uncertainty compounds over longer windows. Ensemble methods failed to offset this degradation, showing that combining weakly predictive models does not necessarily yield stronger performance when the base signals themselves are weak.

The **simple ensembles**, averaging and voting regressors, showed moderate improvement over weaker tree-based models but never surpassed Ridge Regression. Their nearly identical scores suggest that averaging and voting contributed similar smoothing effects. The **stacking ensemble**, which used Ridge as a meta-learner, performed poorly for short horizons (3- and 6-month forecasts) and only marginally improved for the 9-month case, though it still remained worse than Ridge alone. This underperformance stems from **model redundancy**, since the base learners already capture overlapping patterns, stacking could not extract new information.

Overall, the ensemble experiments reaffirmed three critical insights. First, **Ridge Regression's regularization** provides the best trade-off between simplicity and robustness, outperforming complex non-linear models in noisy macroeconomic data. Second, **ensemble diversity** is limited when all base models rely on similar features, which restricts the ensemble's ability to add predictive value. Third, the consistently **negative R^2 values** reflect the reality of efficient markets and random-walk–like behavior in aggregate economic indicators, where historical rates contain little forward-looking information.

Thus, ensemble learning did not improve upon Ridge Regression. The results emphasize that in economic forecasting, where predictability is inherently constrained, **simpler regularized linear models** often outperform more complex alternatives. Future work should prioritize improving feature quality, such as incorporating external macroeconomic indicators, sentiment variables, or time-series–specific methods like ARIMA or VAR, rather than increasing model complexity.

6) Conclusion

This project systematically explored how financial market indicators, such as Treasury yields, commercial paper rates, and yield spreads, relate to future economic activity using machine learning techniques. Through a sequence of exploratory analysis, feature engineering, and multi-model evaluation, the study assessed predictive performance across several forecasting horizons, aiming to bridge economic theory with modern data-driven methods.

From the analyses, several consistent insights emerged. First, regularized linear models such as Ridge Regressionconsistently provided the most stable and interpretable forecasts, outperforming more complex tree-based algorithms and ensemble approaches. This reinforces the principle that in macroeconomic contexts characterized by high noise-to-signal ratios and structural uncertainty, simplicity and regularization often yield more reliable generalization than highly flexible models.

Second, the negative R^2 values across all horizons highlight the intrinsic difficulty of economic forecasting. Market efficiency, rapid information diffusion, and non-stationary dynamics collectively constrain the predictive power of models based solely on historical indicators. These results align with established economic theory, suggesting that financial markets incorporate available information almost instantaneously, leaving limited room for mechanical prediction.

Third, the comparative results across the 3-month, 6-month, and 9-month horizons demonstrated a clear deterioration in predictive accuracy over longer timeframes, emphasizing that economic relationships weaken as uncertainty compounds over time. Despite extensive feature engineering, including yield-curve slopes, credit spreads, and volatility measures, the models could not capture consistent predictive structure beyond short-term horizons.

In practical terms, these findings suggest that future efforts should focus less on algorithmic complexity and more on enhanced feature design, inclusion of exogenous macroeconomic indicators, and time-series–specific modeling frameworks such as ARIMA, VAR, or hybrid econometric–ML systems. Incorporating regime-switching mechanisms or sentiment-driven variables could also improve responsiveness to cyclical turning points.

Ultimately, this project demonstrates both the potential and the limits of machine learning in economic forecasting. While data-driven models can reveal valuable patterns and validate theoretical expectations, they remain constrained by the stochastic nature of macroeconomic systems. By grounding model design in economic logic and emphasizing interpretability over pure performance, future research can better integrate machine learning insights into policy formulation, investment strategy, and risk management.

References

Textbooks:

1. Raschka, Sebastian, Yuxi Hayden Liu, and Vahid Mirjalili. *Machine Learning with PyTorch and Scikit-Learn.* 4th Edition. Packt Publishing, 2022.
2. Bowles, Michael. *Machine Learning in Python.* Wiley, 2015.
3. Hackeling, Gavin. *Mastering Machine Learning with Scikit-Learn.* 2nd Edition. Packt Publishing, 2017.
4. Alpaydin, Ethem. *Introduction to Machine Learning.* The MIT Press, 2014.
5. Moncecchi, Guillermo, and Raúl Garreta. *Learning Scikit-Learn: Machine Learning in Python.* Packt Publishing, 2013.

Google Colab Notebook Link

Colab Notebook:
https://colab.research.google.com/drive/1X2vUVd5OLgPqdMy8D7NJ_ll2h4jO2pfl?usp=sharing